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All-dielectric meta-waveguides for on-chip integration

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Meta-waveguides have demonstrated unprecedented capabilities for controlling guided waves in both free space and on-chip. However, the designs for on-chip guided waves control are still insufficient, with only a few functions being proposed. Moreover, some designs use metal antennas to provide the necessary phase shifts, which introduce undesirable Ohmic losses and thermal effects. Here, we demonstrate a set of silicon nitride (Si_3N_4) phase-gradient meta-waveguides based on all-dielectric silicon (Si) antennas. The designed meta-waveguide reflector achieves robust, broadband, high-efficiency, and high-fidelity reflection of the TE_{00} mode. The designed meta-waveguide Fano resonator achieves Fano resonance over a broad wavelength range. Additionally, the designed meta-waveguide Fabry-Pérot (FP) cavity supports compact on-chip optical sensing and information processing. The designed meta-waveguide multimode switch enables actively tunable multimode-selective output. These all-dielectric meta-waveguides for on-chip guided waves control will expand the component library of photonic integrated circuits (PICs) and provide positive demonstrations for high-density integrated on-chip devices.

The ability to generate, transmit, process, and receive optical signals in a smaller-scale device is crucial for numerous applications in nano-photonics and integrated photonics. Photonic integrated circuits (PICs) enable these by integrating various light-controlling components onto a single chip, which offers an attractive platform for optical information processing^{1–4}, integrated quantum optics^{5,6}, on-chip chemical and biological sensing^{7,8}. However, the latest advancements in PICs are hindered by the limitations in its most fundamental building block of traditional optical waveguides⁹. Achieving more effective manipulation of guided waves and overcoming the inherent limitation of waveguides being used solely for light transmission remains a challenge. Therefore, there is an urgent need for novel form waveguide designs that enable compact and multifunctional on-chip devices.

Recently, emerging compact meta-waveguide devices have paved a new path for extended functionalities and improved performance of photonic integrated devices¹⁰. It has been demonstrated that arranging sub-wavelength meta-atoms (antennas) on optical waveguides to act as Mie resonators can control guided waves both in free space and on-chip¹¹. Although the concept of meta-optics has been adapted for designing meta-waveguide devices, current research mainly focuses on manipulating guided waves in free space, such as beam deflection and focusing^{11–14}, polarization and wavelength demultiplexing^{15,16}, optical holography^{14,17,18}, wavefront shaping¹⁹, orbital angular momentum generation¹¹, and particle manipulation²⁰. Such functionalized meta-waveguide devices typically rely on the far-field radiation of meta scatterers into free-space. As a result, their

conversion efficiency is inherently constrained by the highly localized light within the waveguides. Moreover, coupling between free-space light and guided waves^{21–24} in meta-waveguide devices also presents the same challenge due to light leakage into the substrates⁹.

Compared to meta-waveguide devices that use Mie resonators to control guided waves in free space, the exploration of meta-waveguides for on-chip guided wave control remains insufficient. Pioneering work by Li et al. demonstrated on-chip mode conversion and asymmetric optical power transmission using phase-gradient meta-waveguides²⁵. Since then, meta-waveguides based on mode conversion have enabled some advancements, including phase-matching-free nonlinear wavelength converters²⁶, mode interference²⁷, and programmable phase-change meta-waveguide mode converters²⁸. However, new functional devices based on meta-waveguide for on-chip guided wave control still need further development. Furthermore, some meta-waveguide designs use metal antennas to provide the necessary phase shifts, which introduces coupling between guided modes and surface waves, result in the optical power dissipated via Joule heating²⁵. Such plasmonic meta-waveguides will introduce undesirable Ohmic losses and thermal effects that significantly impact the performance of on-chip devices.

Here, a series of silicon nitride (Si_3N_4) meta-waveguide devices that based on all-dielectric silicon (Si) antennas were numerically demonstrated. We proposed a compact meta-waveguide reflector that operates independently of waveguide width and achieves broadband, high-efficiency and

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high-fidelity reflection of the TE_{00} fundamental mode. Additionally, incomplete mode conversion and interference within all-dielectric meta-waveguides are employed to design an compact Fano resonator. We also designed all-dielectric meta-waveguides for compact on-chip Fabry-Pérot (FP) cavity and tunable multimode switch. All these devices are presented for the first time and can be used for PICs as standard components. This work further demonstrates the ability of meta-waveguides to control guided waves on-chip, offering valuable insights for the development of high-density integrated devices in PICs.

Results

Fundamentals and design principles

It is possible to effectively control the amplitude, phase, and polarization of guided waves in a meta-waveguide by constructing an optical field overlap between the guided mode and the meta-atom scattering field at the cross-section²⁷. Figure 1a–c illustrates the schematic of a phase-gradient meta-waveguide for controlling guided waves on-chip. A series of antennas (meta-atoms) functions as Mie resonators, which are arranged with a fixed period dx on the surface of a Si_3N_4 waveguide. These continuous scatterer antennas have the same width and height but with lengths decreasing (or increasing) in the y -direction, which introduce a unidirectional phase gradient $d\phi/dx$ along the negative x -direction.

For a mode conversion process, the propagation constant of the guided wave changes from the input mode β_{in} to the output mode β_{out} , which can be expressed as $\beta_{in} - \beta_{out} = k_0(n_{in} - n_{out}) = N \cdot d\phi/dx$, where $k_0 = 2\pi/\lambda$ is the free-space wavevector, n_{in} and n_{out} are the effective mode indexes of the input and output modes, respectively. N is the number of interactions between the guided mode and metasurface (antenna array)^{25,27,28}. To construct the metasurface, we arrange a set of antennas with varying lengths into an array with a subwavelength spacing of dx between adjacent ones, and set the phase response between two adjacent antennas to be $d\phi$. Thus a constant phase gradient of $d\phi/dx$ along the gradient metasurface is created. As shown in

Fig. 1a, when the guided mode (e.g., TE_{00}) is incident from the left port along the positive x -direction (against the phase gradient direction afforded by the Si or Au antennas), i.e., the forward direction, the effective mode index and the total internal reflection angle decrease. Consequently, the TE_{00} mode converts into a higher-order mode (e.g., TM_{10}) and is output from the right port. Similarly, according to the Lorentz reciprocity theorem, a higher-order waveguide mode (TM_{10}) incident from the right port along the negative x -direction can also be efficiently converted to a lower-order mode (TE_{00}) at the left port. However, when the propagating mode along the phase gradient direction afforded by the antennas is the TE_{00} fundamental mode, i.e., backward direction, it becomes an attenuated mode and cannot propagate within the waveguide.

References 25,29 have proposed a metal antenna-based meta-waveguide scheme, as shown in Fig. 1b. They use Au antennas to provide the required phase gradient; however, when the TE_{00} fundamental mode is incident from the backward direction, the TE_{00} fundamental mode is ultimately coupled into surface waves and dissipated through Joule heating via the plasmonic metasurface²⁵. The use of metal antennas introduces Ohmic losses and thermal effects, which are undesirable for on-chip integrated devices and also pose compatibility issues with complementary metal-oxide-semiconductor (CMOS) processes¹². Moreover, the low efficiency of metal meta-waveguides remains a challenge in practical applications that needs to be emphasized.

Here, we are the first to demonstrated that when using all-dielectric Si antennas to provide the required phase gradient, the backward incidence of the TE_{00} fundamental mode does not couple into surface waves but is instead broadband, high-efficiency and high-fidelity reflected, as shown in Fig. 1c.

As previously mentioned, a prerequisite for achieving such guided mode control is to ensure that the scattering near-field of the meta-atom sufficiently overlaps with the modal field on the waveguide cross-section, which can be expressed as^{25,27}:

$$\frac{|\iint \mathbf{E}_{\text{antenna}}(y, z) \cdot \mathbf{E}_{\text{mode}}^*(y, z) dy dz|^2}{\left(\iint |\mathbf{E}_{\text{antenna}}(y, z)|^2 dy dz\right) \cdot \left(\iint |\mathbf{E}_{\text{mode}}(y, z)|^2 dy dz\right)}, \quad (1)$$

where $\mathbf{E}_{\text{antenna}}(y, z)$ represents the near-field scattered light around an individual antenna, $\mathbf{E}_{\text{mode}}(y, z)$ represents the transverse cross-sectional field of the light propagating in the waveguide, and the integration is over the transverse waveguide cross section. By appropriately selecting the size and orientation of antennas on the waveguide surface, one can control the overlap described by Eq. (1), thereby fine-tuning the coupling efficiency between the two waveguide modes. This implies that the dielectric antennas must be meticulously designed. To this end, we employed the full-vector finite-difference time-domain (FDTD) method to simulate the scattering phase response of an individual Si antenna on the Si_3N_4 waveguide. Detailed simulation design can be found in Section 1 of the Supplementary Information.

Compact meta-waveguide reflector

Reflectors play a pivotal role in PICs, finding wide applications in attenuators, filters, laser cavities, and other fields³⁰. Currently, there are various types of on-chip reflectors, including loops based on multimode interferometers (MMIs) or directional couplers (DCs), distributed Bragg reflectors (DBRs), high-contrast gratings (HCGs), metal-coated reflectors, and inverse-designed reflectors^{30–32}. However, loop structures are constrained by scattering losses from Y-junctions or MMI components, and the use of DCs faces challenges such as low fabrication tolerance and limited optical bandwidth. In comparison, DBRs, as the most common on-chip reflectors, can be realized through sidewall corrugations, embedded nano-holes, and deep trenches. Nevertheless, such invasive etching introduces significant diffraction losses³². While HCGs show promise, further size reduction and efficiency improvements are required³⁰. Metal-coated reflectors and inverse-designed reflectors are hindered by ohmic losses and inherent high reflection losses due to stringent feature sizes, respectively³². In summary, despite the advancements in integrated

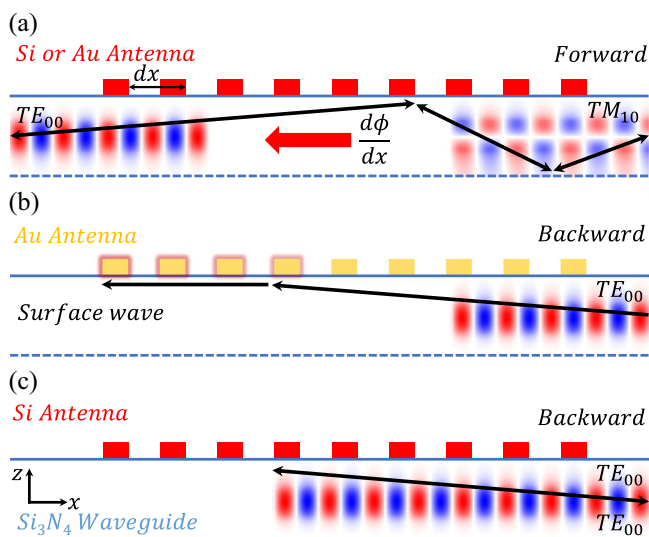


Fig. 1 | Schematic of phase-gradient meta-waveguides, where the unidirectional phase gradient $d\phi/dx$ can be introduced by all-dielectric Si or metallic Au. **a** In the forward direction, both Si and Au antennas phase-gradient meta-waveguides can convert the incident TE_{00} fundamental mode into a higher-order mode (e.g., TM_{10}) through the continuous interaction with antennas, and vice versa. **b** In the backward direction, the metallic Au antennas phase-gradient meta-waveguide couples the incident TE_{00} fundamental mode into a surface wave that propagates along the plasmonic gradient antennas, resulting in complete absorption of optical power. **c** In the backward direction, the all-dielectric Si antennas phase-gradient meta-waveguide perfectly reflects the TE_{00} fundamental waveguide mode with high efficiency and fidelity.

reflectors, there remains a pressing need for ultra-compact, low-loss, and robust solutions in this field.

Previous works^{25,29} demonstrated that metallic antenna-based phase-gradient meta-waveguides can achieve coupling between the TE_{00} mode and surface waves under backward incidence, resulting in the optical power dissipation through Joule heating. This mechanism was employed to design meta-waveguide devices for asymmetric optical power transmission. However, such dissipation does not occur when using all-dielectric phase-gradient antennas. As mentioned earlier, we were the first to discover that all-dielectric phase-gradient meta-waveguides enable broadband, high-efficiency and high-fidelity reflection of TE_{00} fundamental mode under backward incidence. This discovery led to the realization of a compact on-chip meta-waveguide reflector, representing an entirely new class of on-chip reflector structures. Here, the compact meta-waveguide reflector is designed to work around 1550 nm. It is based on a multimode Si_3N_4 waveguide with a height of 800 nm and a width of 1200 nm, as detailed in Table S1, Section 2 of the Supplementary Information.

In the forward direction, the incident TE_{00} mode gradually diminishes (Fig. 2b1) and converts into the TM_{10} mode (Fig. 2b2). In the backward direction, the incident TE_{00} mode maintains perfect high-fidelity reflection (Fig. 2c1), with only a minimal TM_{10} higher-order mode present due to the mode conversion from the reflected TE_{00} fundamental mode during forward propagation (Fig. 2c2). This has negligible impact on fidelity and can be mitigated using single-mode waveguide designs. The distribution of Poynting vector displayed in Fig. 2b3, c3 highlights the continually strong interaction between the all-dielectric antennas and the propagation TE_{00} fundamental mode, indicating limited light scattering into free space and the substrate. In Fig. 2b4, c4, S_f and S_b are the positions of the mode light source under forward incidence and backward incidence, respectively. Figure 2b4 shows that mode conversion results in some reflection, with limited light returning to the left port. Figure 2c4 demonstrates effective mode reflection, with nearly no transmission through the left port.

To illustrate this further, we plotted the transmission and reflection coefficients for both forward and backward incidence at the left and right ports in Fig. 2d. In the forward direction, the transmission at the right port of the Si-based meta-waveguide reaches approximately 80% over a broad wavelength range during forward incidence (approximately 75% in TM_{10} mode and less than 0.4% in TE_{00} mode), roughly double that of Au-based antennas, accompanied by some mixed-mode reflection of TE_{00} and TM_{10} back to the left port. In contrast, under backward incidence, the reflection at the right port approaches 1, while that at the left port nears 0, confirming the efficient broadband reflective characteristics of our designed reflector. This design can also be utilized for asymmetric optical power transmission in waveguides; however, it is important to clarify that such asymmetrical transmission, based on mode coupling and mode reflection, is reciprocal and distinct from that of optical isolators^{25,33,34}.

Compared to phase-gradient meta-waveguides utilizing metal antennas, the all-dielectric Si antenna meta-waveguides exhibits lower losses and, crucially, achieves broadband, high-efficiency and high-fidelity reflection of TE_{00} mode. By maintaining the antenna parameters fixed, we varied the waveguide width and recorded its reflection efficiency, as shown in Fig. 2e. The results indicate that the meta-waveguide reflector possesses remarkable robustness, operating independently of waveguide width and tolerating significant fabrication variations. Additionally, we analyzed the impact of manufacturing errors in the antenna's geometric dimensions on the performance of the meta-waveguide reflector in Section 4 of the Supplementary Information. Even with a fabrication error of up to ± 10 nm, our device maintains excellent performance, with a reflectivity greater than 95%. This indicates that our design exhibits strong fabrication tolerance, ensuring robust performance under realistic manufacturing conditions. Notably, although the meta-waveguide reflector exhibits broadband characteristics, significant deviations from the center operating wavelength can result in inefficient mode conversion and reflection. To accommodate different center operating wavelengths, it is necessary to re-simulate the antenna's phase response at the new center wavelength, as well as re-select the appropriate phase gradient and antenna lengths.

Compact meta-waveguide Fano resonator

Devices such as optical switches, modulators, sensors, and isolators developed based on the Fano resonance of the waveguide structures have led to significant advancements in nano-photonics³⁵. Initially, Fano resonance was typically realized in plasmonic waveguide structures; however, thermal losses in these metallic materials constrained the performance of optical devices. To address this issue, researchers have achieved Fano resonance in dielectric waveguides through three methods: photonic crystal nanobeam cavities (PCNCs)^{36,37}, microring resonators (MRRs)^{2,38}, and Mach-Zehnder interferometer (MZIs) structures³⁹. PCNCs consist of line-defect photonic crystal dielectric waveguide, but the etched air holes of these defects impose significant limitations on applications in biochemical fields and demand high fabrication precision. Fano resonators constructed using MRRs and MZIs typically introduce phase differences or coupling to achieve Fano resonance, yet they still face challenges such as high insertion loss and larger footprint. Notably, among all resonance modes in the transmission spectra of some on-chip waveguide Fano devices, only a portion exhibits Fano line shapes. Given the importance of practical applications, exploring methods to achieve full-wavelength Fano resonance spectra is crucial.

To address this, we first demonstrate a Fano resonator based on mode interference in all-dielectric meta-waveguides, which exhibits Fano resonance over a broadband wavelength range. This design features an ultra-compact structure that significantly reduces the footprint, offering a novel Fano resonance configuration that could inspire new developments in the field of PICs. The schematic of the compact meta-waveguide Fano resonator is shown in Fig. 3a. It is designed based on a multimode Si_3N_4 waveguide with a height of 800 nm and a width of 1000 nm, consisting of two oppositely placed phase-gradient Si antennas, as detailed in Table S2, Section 2 of Supplementary Information.

As previously indicated, the all-dielectric phase-gradient meta-waveguide can convert TE_{00} mode into TM_{10} high-order mode in forward propagation. In backward propagation, we found that it achieves robust broadband reflection of the TE_{00} mode light. To this end, we meticulously configured the meta-waveguide so that the TE_{00} mode propagating forward is not completely converted into the TM_{10} mode upon encountering the first phase-gradient Si antenna array (achieved by reducing the number of antennas at the same phase gradient, which decreases the mode conversion rate, resulting in TE_{00} mode not being completely converted to the TM_{10} mode); instead, the unconverted TE_{00} mode light is reflected back and forth between the two phase-gradient antenna arrays upon reaching the second phase-gradient Si antenna array. Here, the transmitted TM_{10} mode acts as a continuous state, while the resonating TE_{00} mode acts as a discrete state, with their interference forming the Fano resonance in the meta-waveguide. Figure 3b1, b2 illustrate this clearly: at the resonant wavelength, a discrete resonating TE_{00} mode exists within the waveguide, significantly increasing the field intensity (Fig. 3b1, E_y), while at non-resonant wavelengths, there is almost no resonating TE_{00} mode (Fig. 3b2, E_y), and the continuously transmitted TM_{10} mode light persists (Fig. 3b1, b2, E_z). We plotted the transmission at the right port, as shown in Fig. 3c, and clearly observed that the designed compact all-dielectric meta-waveguide exhibits Fano resonance lineshapes across a broad wavelength range. We extracted one of the Fano resonant peak and fitted its quality factor (Q -factor), which was approximately 4628. Notably, the Q -factor decreases significantly with increasing resonant wavelength, attributed to the intensity and phase changes required for mode interference. Manufacturing errors in the antenna's geometric dimensions can also lead to a decrease in the Q -factor, as discussed in Section 4 of the Supplementary Information. In summary, we have demonstrated an ultra-compact all-dielectric meta-waveguide Fano resonator over a broadband wavelength range, characterized by a simple and compact structure with a commendable Q -factor, making it suitable for widespread applications in optical information processing and sensing within PICs.

Meta-waveguide FP cavity

The monolithic integration of FP cavities has many applications in sensing and optical communication, including label-free sensing, high-finesse

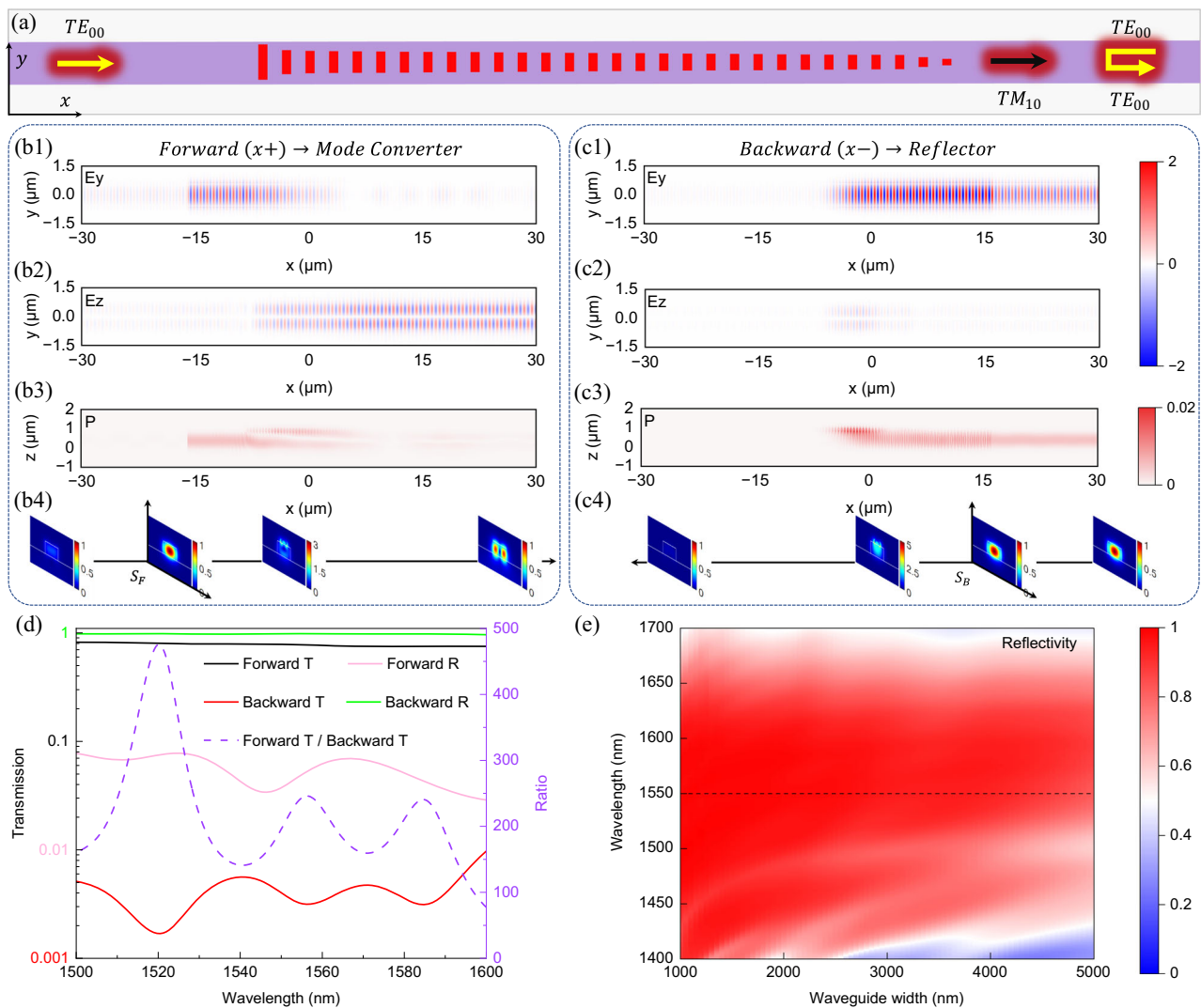


Fig. 2 | The designed meta-waveguide reflector. **a** Top-view schematic of the compact all-dielectric phase-gradient meta-waveguide. **b1–b4** The all-dielectric phase-gradient meta-waveguide functioning as a mode converter during forward incidence, converting TE_{00} fundamental mode light into the higher-order TM_{10} mode: distribution of the E_y field (**b1**), E_z field (**b2**), Poynting vector P (**b3**) and E field (**b4**) at a wavelength of 1550 nm. **c1–c4** The all-dielectric phase-gradient

meta-waveguide functioning as a reflector for TE_{00} fundamental mode light during backward incidence: distribution of the E_y field (**c1**), E_z field (**c2**), Poynting vector P (**c3**) and E field (**c4**) at a wavelength of 1550 nm. **d** Transmission and reflection coefficients at the left and right ports during forward and backward incidence. **e** The effect of waveguide width on reflectivity at different wavelengths.

filtering, semiconductor lasers, and frequency comb generation^{32,40}. Chip-level FP cavities offer greater design flexibility, smaller footprint, and higher integration density compared to whispering-gallery-mode microring cavities. Moreover, FP structures can support multiple transverse modes over a broad spectrum and are not limited by other physical rules in terms of the frequency spacing between the transverse modes, which is crucial for optimizing quantum and nonlinear optical processes on-chip⁶.

Here, we demonstrate an FP cavity based on all-dielectric meta-waveguide structure by coupling an optimized meta-waveguide reflector to an input-output waveguide, as shown in Fig. 4a. The meta-waveguide FP cavity is designed using a multimode silicon nitride waveguide with a height of 800 nm and a width of 1000 nm, with antenna selections detailed in Table S3, Section 2 of Supplementary Information. The E_y field intensity at resonant and non-resonant conditions is shown in Fig. 4b, indicating that the meta-waveguide FP cavity operates based on the reflection resonance of the TE_{00} mode. Besides, the meta-waveguide FP cavity exhibits a Lorentzian resonance line shape over a broad wavelength range, as illustrated in Fig. 4c. We extracted one of the resonance peaks and the fitted quality factor is about 7954. Notably, significant manufacturing errors in the antenna's geometric

dimensions will lead to a decrease in the Q-factor, as discussed in Section 4 of the Supplementary Information. This novel meta-waveguide FP cavity structure will significantly advance the development of on-chip integrated FP cavities, enriching the silicon photonics toolbox and paving the way for numerous applications.

Meta-waveguide multimode switch

Multimode switch is a key component of mode-division multiplexing (MDM) technology in on-chip optical communication, optical signal processing and optical interconnects. However, compared to single-mode optical switches, conventional multimode switches still remain a significant challenge in PICs. These challenges primarily arise from limitations such as modal crosstalk, integration compatibility, and the capacity to support multiple modes simultaneously. To address this issue, we designed a multimode optical switch that integrates meta-waveguides with a Mach-Zehnder Interferometer (MZI) network, as illustrated in Fig. 5a. The proposed switching scheme can be extended for more modes and can effectively improve the system transmission capacity.

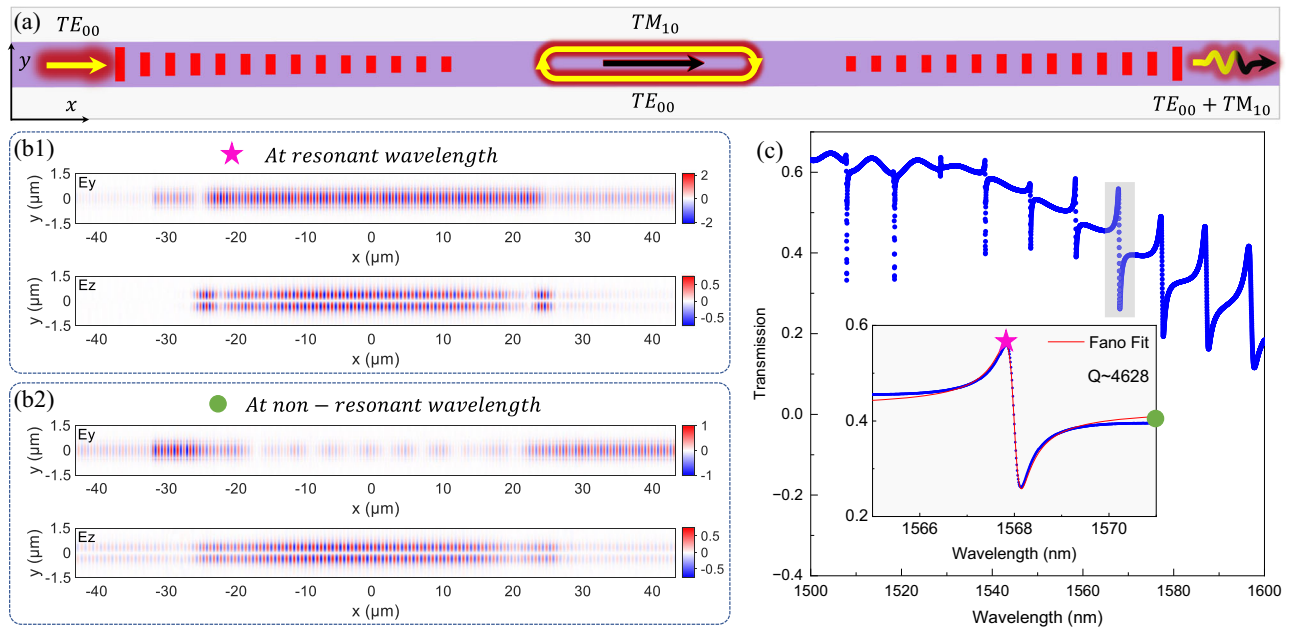


Fig. 3 | The designed meta-waveguide Fano resonator. **a** Top-view schematic of the compact all-dielectric meta-waveguide Fano resonator. **b1** E_y and E_z field distributions of the all-dielectric meta-waveguide at the resonant wavelength (1567.83 nm). **b2** E_y and E_z field distributions of the all-dielectric meta-waveguide at the non-resonant wavelength (1571.24 nm). **c** Transmission spectrum at the right port, with the inset showing the Fano resonance fitting at one of the resonant wavelengths.

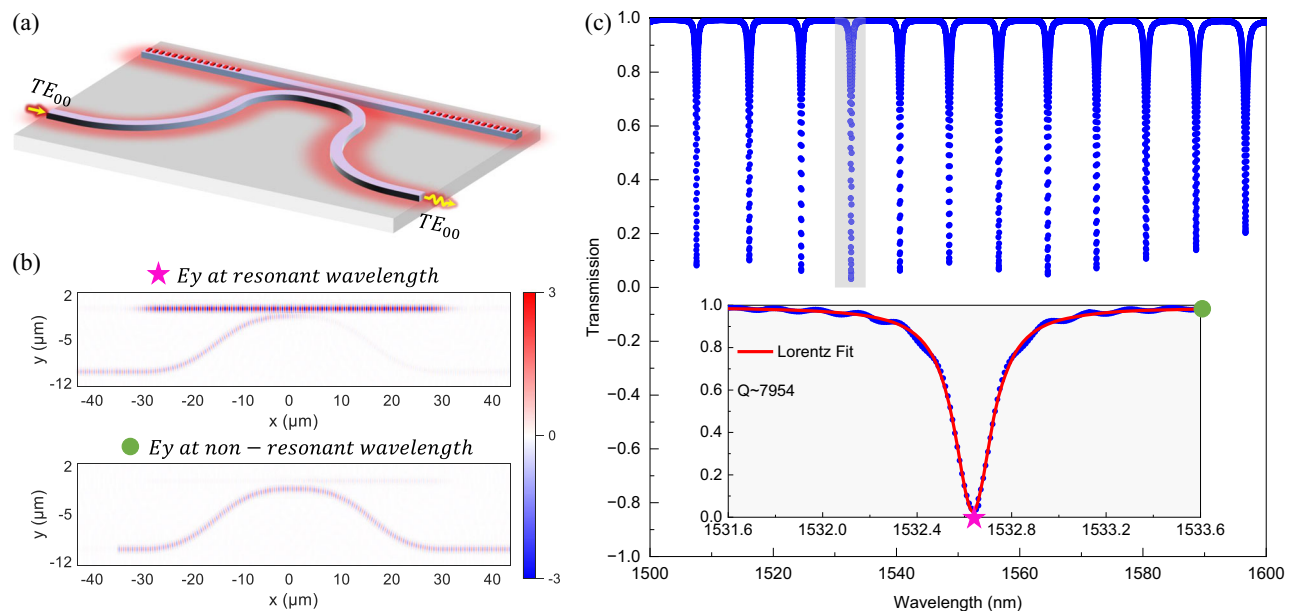


Fig. 4 | The designed meta-waveguide FP cavity. **a** Schematic of the all-dielectric meta-waveguide FP cavity. **b** E_y field distributions of the dielectric meta-waveguide at resonant (1532.63 nm) and non-resonant wavelengths (1533.60 nm). **c** Transmission spectrum of the right port, with the inset showing the Lorentzian fitting at one of the resonant wavelengths.

The different arrangements of antennas will cause the TE_{00} fundamental mode to be converted into different modes²⁵. Here, as a conceptual demonstration, we demonstrate a three-port multimode switch, where the TE_{00} mode can be switched to TM_{10} , TM_{00} , or maintained as TE_{00} output. The antenna arrangements of the meta-waveguide mode converter is illustrated in Fig. 5b, with the antennas selection described in Table S4, Section 2 of Supplementary Information. The designed meta-waveguide multimode switch is constructed using unit MZI structure (red dashed box in Fig. 5a), enabling output selection at a specified wavelength by controlling the phase shift between the two arms of the unit MZI structure. In practice, the phase shift between the two arms of the unit MZI can be actively tuned by integrated heater modules for thermo-optic tuning, as shown by the yellow

electrical pads in Fig. 5a. The output spectra from the upper (P_1) and lower (P_2) ports of the unit MZI structure in the red dashed box of Fig. 5a are shown in Fig. 5c. This allows the incident TE_{00} mode to be assigned to any output port, and the output TE_{00} mode from a selected port is subsequently converted to the corresponding mode through the meta-waveguide mode converter. We simulated the E-field at the output ports of the meta-waveguide at a wavelength of 1755 nm in Fig. 5c, confirming that the results align well with expectations, demonstrating the mode-switching functionality. Additionally, the proposed meta-waveguide multimode switch is highly scalable. By increasing the number of MZI units, it is possible to control the output modes of more ports, thereby enhancing its application flexibility and on-chip integration potential.

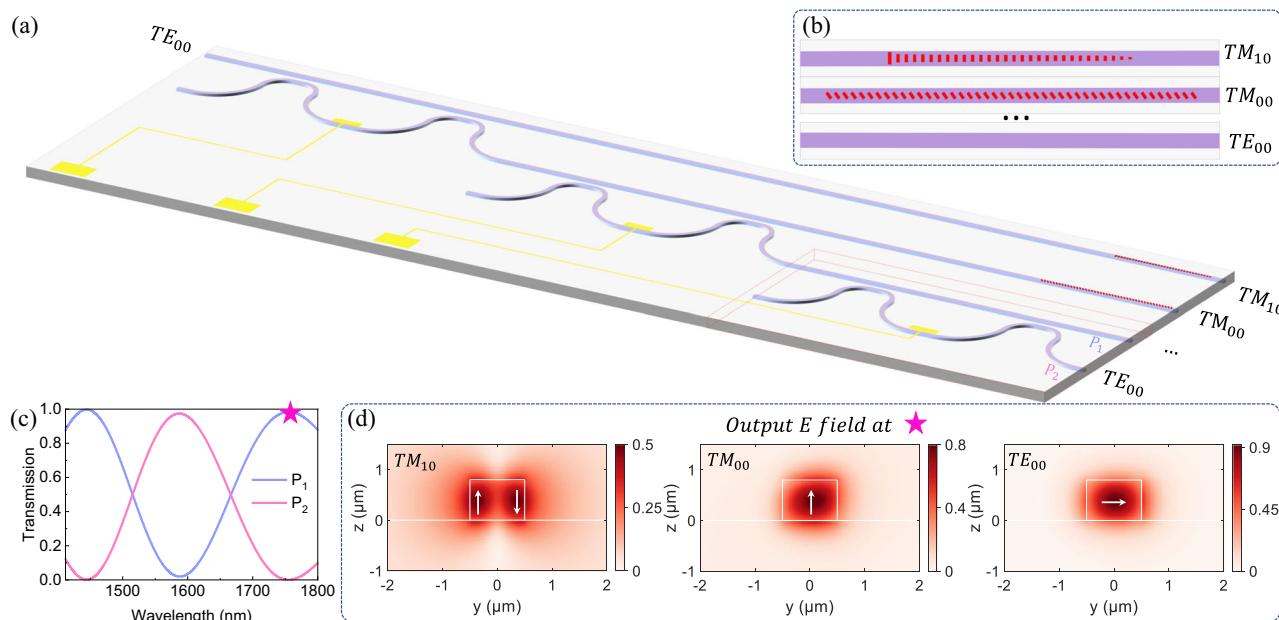


Fig. 5 | The designed meta-waveguide multimode switch. **a** Schematic of the tunable all-dielectric meta-waveguide multimode switch. **b** Antenna arrangement required for the meta-waveguide converters that converts the TE_{00} fundamental mode into different modes. **c** Transmission spectrum of the unit MZI structure. **d** Output E -field distribution of the three-port multimode switch at a wavelength of 1755 nm.

Discussion

In conclusion, we have designed a series of high-density on-chip integrated devices based on all-dielectric Si phase-gradient meta-waveguides, including reflector, Fano resonator, FP cavity, and multimode switch. To highlight the advantages of these proposed meta-waveguide devices, we provide a comparative discussion on the performance of these designed devices with current research in Section 3 of Supplementary Information. Unlike the current on-chip reflector that rely on coupling, etching, or inverse design, our meta-waveguide reflector achieves robust and near-perfect reflection of TE_{00} mode light over a wide bandwidth (>200 nm) with a reflectivity of approximately 97%. The meta-waveguide Fano resonator we designed ingeniously leverages the interference between the discrete resonant TE_{00} mode and the continuous propagation TM_{10} mode within the meta-waveguide, realizing Fano resonance over a broad spectral range with a promising Q -factor suitable for on-chip sensing applications. Similarly, the proposed meta-waveguide FP cavity is also viable for on-chip sensing and can function as a competitive candidate of conventional microring resonators for on-chip information processing. Finally, we introduced a scalable meta-waveguide multimode switch based on the MZI network, which shows potential for application in MDM. While our study is based on the simulations, current fabrication technologies are already capable of processing various Si antenna array structures on Si_3N_4 waveguide platforms^{25,41–43}, indicating that the designed all-dielectric meta-waveguide devices are feasible for actual fabrication. Furthermore, we have discussed the impact of fabrication errors on meta-waveguide device performance in Section 4 of the Supplementary Information. Overall, the results show that the designed meta-waveguides demonstrate strong tolerance to manufacturing errors, ensuring robust performance under practical fabrication conditions. In contrast to meta-waveguide devices designed for free-space optical modulation, our research focuses on controlling guided waves on-chip. The proposed all-dielectric meta-waveguide devices are highly compact and show significant potential as standard components for PICs, particularly for the applications in on-chip optical information processing and sensing.

Data availability

No datasets were generated or analysed during the current study.

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Author contributions

J.W. conceived the idea, conducted the simulations, designed all the devices, and wrote the manuscript. W.W. assisted with the simulations and revised the primary manuscript. W.D., J.L., P.J., S.L. and K.Z. discussed this work and contributed to the drafting and revision of the manuscript. K.Z. supervised the project, reviewed the manuscript. All authors read and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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